

Abdurbaki Sanlan

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Summary

Currently pursuing an PhD. in Aerospace Engineering at Istanbul Technical University and conducting research at the ITU Aerospace Research Center. Experienced in the design and implementation of advanced GNC algorithms for aerospace applications, with a strong focus on developing and applying novel approaches to improve the autonomy, precision, and robustness of aerial systems.

Education

M.Sc. in Aerospace Engineering <i>Istanbul Technical University, Istanbul, Turkey</i> GPA: 3.78/4.00	2024 – 2025
B.Sc. in Aeronautical Engineering <i>Istanbul Technical University, Istanbul, Turkey</i> GPA: 3.36/4.00	2019 – 2024
High School <i>Ergun Oner Mehmet Oner Anadolu Lisesi, Turkey</i>	2016 – 2019

Professional Experience

Researcher ITU Aerospace Research Center, Istanbul, Turkey - GPS/GNSS Denied Navigation - Implemented a terrain-matching solution for 6-DOF aircraft state estimation by fusing INS data and point-cloud measurements through the Unscented Kalman Filter (UKF) and Particle Filter (PF). - Developed a visual navigation-based UAV positioning pipeline that integrates Visual Inertial Odometry (VIO) with Generative AI (GANs)–enabled domain adaptation of satellite imagery matching within a Particle Filter (PF) framework. - Anti Skid Breaking System Design - Developed an optimal friction curve estimation model using Particle Swarm Optimization (PSO), integrating a hybrid approach with a Sliding Mode Estimator (SME) for enhanced performance. - Conducted research on effective radius estimation for systems operating under measurement-deficient conditions.	Nov 2023 – Present
Engineer Intern Baykar Teknoloji, Istanbul, Turkey Worked on swarm UAV formation flight algorithms based on Artificial Potential Field.	Jul 2023 – Sep 2023
Engineer Intern TUSAŞ, Istanbul, Turkey - Assisted in trim and linearization for a 6-DOF Aircraft model - Learning visualization of aircraft model on Simulink/Flight Gear.	Oct 2022 – Mar 2023

Skills

- 🔹 **Advanced:** Python, MATLAB, LaTeX
- 🔹 **Intermediate:** ROS, Simulink, Git, C++

Languages

- English (B2, IELTS 7/9)
- Turkish (Native)

Publications

- A.B. Sanlan, H. T. Bagci, E. Koyuncu** "Particle Filter-Based Localization Using Visual Feature Synchronization in GNSS-Denied Navigation" 2025
The 25th Integrated Communications, Navigation and Surveillance Conference, Brussels, Belgium.
- A.B. Sanlan, F. Erol, M. Abu-Khalaf, E. Koyuncu** "Terrain-Aided Navigation Using a Point Cloud Measurement Sensor" 2025
arXiv e-prints, abs/2510.06470.

Projects

Machine Learning:

- Designed and trained a CycleGAN-based generative model for image-to-image translation, effectively minimizing domain discrepancies across diverse datasets
- Developed and trained a deep learning pipeline for automated species identification using audio recordings, leveraging signal processing and neural network techniques.
- Implemented a grounded image-classification framework built on the Segment Anything Model (SAM), enhancing segmentation precision and object recognition accuracy.

Optimal Control: Finding the trajectory that will take a commercial plane from point A to point B with minimum fuel consumption and minimum time using single shooting, multiple shooting and direct collocation optimal control methods.

Reinforcement Learning: Finding the trajectory that will take a passenger commercial plane from point A to point B with minimum fuel consumption and minimum time with Reinforcement Learning methods, and finding the trajectory that avoids fixed obstacles in a two-dimensional environment and reaches the target.

Dynamic Programming: Modeled one-on-one 3D air combat scenarios and developed sub-optimal dogfighting algorithms using Approximate Dynamic Programming to achieve effective tactical maneuvering under real-time computational constraints.

LQR & LQI Controller: Designed controllers for the F-16 model (Aerobench-vv) using MATLAB/Simulink.

Inverted Pendulum RL Control: Implemented reinforcement learning-based control for an inverted pendulum system.

Volunteer Experience

Software Team Lead on University UAV Team ITU SKY Sep 2022 - Apr 2024
ITU, Turkey

- Developing and leading mission planning, using image processing for object detection-tracking and designing custom PID controllers for *TEKNOFEST* – 2024 competition.

Organizational Coordinator Sep 2022
SAHA EXPO 2022, Istanbul, Turkey

- Coordinated field setup for national and international exhibitors under ITU Dean's Office.

Participant Jun 2022 – Sep 2022
USA Work & Travel Program, Virginia Beach, VA, USA

- Engaged in a three-month cultural exchange program.

Certifications

- Python Advanced Programming (Udemy), Jan 2022
- Exercising Leadership: Foundational Principles (edX), Apr 2022
- PLM Education Certificate (TUSA S), May 2022
- Python for Data Science & ML (Udemy), Aug 2022

- Practical AI with Python & RL (Udemy), Sep 2022

Hobbies

Soccer, Computer Games, Reading Recent Aerospace Research